Project Portfolio and Technical Project Descriptions

1. **Project Title: AI-Based Acoustic Wave Monitoring System for Rail Defect Detection and Predictive Rail Health Monitoring**

Project Description: A low-cost STM32-based embedded AI system that monitors railway track health using vibration and piezoelectric sensors. It processes sensor signals using FFT and machine learning to identify abnormal vibration patterns and provide early warnings of potential rail defects.

2. **Project Title: Battery Redundancy Switching System**

Project Description: An intelligent multi-battery management system that monitors battery parameters through Smart BMS using CAN communication and an STM32 controller. It manages solar/grid charging and automatically switches between healthy battery packs based on SoC and fault conditions to maintain a reliable DC power supply.

3. **Project Title: Smart Industrial Motor Health Monitoring and Protection System**

Project Description: An STM32-based embedded system designed to monitor industrial motor health in real time using voltage, current, temperature, vibration, and RPM sensors. It detects abnormal operating conditions and provides early fault warnings to help prevent motor damage and unexpected downtime.

4. **Project Title: Design and Performance Analysis of a Low-Power Digital System Using Clock Gating on FPGA**

Project Description: Designed and implemented an 8-bit counter-based digital system on the PYNQ-Z2 FPGA to compare baseline, clock-enable, and BUFGCE-based clock-gating techniques. Analysed their functionality, switching activity, power consumption, timing performance, and FPGA resource utilization using Verilog and Xilinx Vivado.

5. **Project Title: Solar Panel Fault Diagnosis Using Virtual Electrical Sensors**

Project Description: An IoT- and edge-AI-based system that monitors solar panel voltage, current, light intensity, and temperature to detect faults such as partial shading, soiling, open circuits, and overheating. It generates virtual electrical parameters to diagnose faults, provide early warnings, and display real-time panel health through a cloud-connected monitoring platform.

6. **Project Title: Vehicle Sensor Interface and Data Acquisition System**

Project Description: Develop a centralized sensor acquisition system for a hybrid two-wheeler that interfaces with multiple vehicle sensors, processes and validates real-time data through an STM32-based HVCU, and communicates vehicle parameters through CAN. The system also supports fault detection, vehicle monitoring, data logging, and integration with the EEE-07 Electronic Vehicle Safety & Protection System.

7. **Smart Desk Assistant Bot Using ESP32-C3 Mini**

Project Description: Developed an ESP32-C3-based smart desk assistant featuring real-time weather updates, NTP-synchronized clock, reminder alerts, and OLED display with remote control through Blynk IoT. Co-authored a research paper presented at an IEEE conference in July 2026.

8. **IoT-Based Smart Energy Meter with Mobile App & Payment System**

Project Description: Developed an ESP32-based smart energy monitoring system integrating voltage and current sensors, RTC, and relay control with a mobile application for real-time energy monitoring and payment processing. Co-authored a research paper published in a UGC Care-approved peer-reviewed journal in December 2025.